SALMAN FARIS ALHAITAMI

Karawang, West Java, Indonesia | +6281296330420 | salmanfarisalhaitami@gmail.com
linkedin.com/in/salman-faris-alhaitami | github.com/SalmanFrs | salman-portfolio-web.vercel.app

PROFESSIONAL SUMMARY

Informatics Engineering graduate from Universitas Komputer Indonesia with a GPA of 3.80/4.00 and a strong foundation in software development, data science, and information technology. Certified Associate Data Scientist (BNSP) with research internship experience at BRIN, involving signal processing, machine learning, and interactive analytical dashboard development. Experienced in using Python, JavaScript, PHP, Java, and SQL, with hands-on experience in web application development using Next.js and React, as well as machine learning and computer vision implementation. Possesses strong analytical thinking, problem-solving, and adaptability skills, with the ability to learn and work with various technologies.

INTERNSHIP EXPERIENCE

Data Science Intern **October 2025 - December 2025**

Badan Riset dan Inovasi Nasional (BRIN) | Bandung, West Java, Indonesia

- Developed an interactive analytics dashboard using Streamlit to analyze earthquake-related infrasound signals and accelerometer data from three BRIN monitoring stations.
- Applied Isolation Forest for anomaly detection, Fast Fourier Transform (FFT) for spectral analysis, and K-Means Clustering with the Elbow Method to group anomalous signals based on their characteristics.
- Implemented an external CSV data processing feature, enabling users to upload and analyze new datasets through preprocessing, anomaly detection, and clustering workflows.
- Deployed the application to a private Hugging Face Spaces repository to support data privacy and functional validation.

EDUCATION

Universitas Komputer Indonesia (UNIKOM) **October 2022 - October 2026**

Bachelor of Informatics Engineering

- GPA: 3.80/4.00
- Thesis: Road Pothole Severity Classification Based on 3D Volume Estimation Using YOLOv11 and PothRGBD Depth Map Fusion

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, PHP, Java, SQL
- **Web Development:** HTML, CSS, React.js, Next.js, Vite, Tailwind CSS, FastAPI
- **Data Science & Machine Learning:** Data Analysis, Data Visualization, Machine Learning, Computer Vision, Natural Language Processing (NLP)
- **Algorithms & Modelling:** YOLOv11-seg, Decision Tree, KNN, SVM, Isolation Forest, K-Means Clustering, Multinomial Naive Bayes, Fast Fourier Transform (FFT)
- **Libraries & Frameworks:** Pandas, NumPy, Scikit-learn, OpenCV, PyTorch, Matplotlib, Seaborn, Streamlit
- **Database & Tools:** MySQL, Git, GitHub, Visual Studio Code, Google Colab, Tableau, Hugging Face, Vercel

CERTIFICATIONS & TRAINING

- **Fundamental Algoritma | Coding Studio**
  Issued: October 2026
- **Gemini Certified Student - University | Google for Education**
  Issued: August 2026 - Expires: August 2029
- **Certified Associate Data Scientist (CADS) | BNSP**
  Issued: March 2026 - Expires: March 2029
- **TOEFL | Language Center, Universitas Komputer Indonesia (Score: 500)**
  Issued: November 2025 - Expires: November 2027
- **Code Generations and Optimization | Hacktiv8 Indonesia**
  Issued: October 2025
- **Code Generation and Optimization Using IBM Granite | IBM**
  Issued: September 2025
- **CCNA: Introduction to Networks | Cisco**
  Issued: February 2024

FEATURED PROJECTS

Pothole Severity Classification | Python, YOLOv11-seg, PyTorch, OpenCV, Streamlit
Research Thesis

- Built a road pothole severity classification system categorizing damage into Low, Medium, and Severe based on 3D volume estimation from RGB images and depth maps using the PothRGBD dataset.
- Implemented YOLOv11-seg for pothole instance segmentation, integrating mask boundaries with depth data to estimate surface area, depth, and total void volume.
- Evaluated model performance on test data, achieving 94.34% Mask Precision, 87.72% Mask Recall, and 86.70% Mask mAP@0.5.

Mini-Blog-IBM | Next.js, TypeScript, Tailwind CSS, Replicate API, IBM Granite
Live Demo | Github Repository

- Developed a web-based blogging application integrated with IBM Granite through Replicate API for AI-powered content generation.
- Implemented frontend functionality using Next.js and TypeScript.
- Implemented Markdown content handling and LocalStorage for locally stored application data.
- Deployed the application using Vercel.

Personal Portfolio Website | React.js, TypeScript, Vite, Tailwind CSS
Live Demo | Github Repository

- Developed a responsive personal portfolio website to showcase professional background, technical skills, certifications, and projects.
- Implemented reusable components and centralized content management using TypeScript.
- Developed interactive project image carousels with automatic sliding, keyboard navigation, and accessibility support.
- Deployed the application using Vercel.

Credit Classification | Python, Scikit-learn, Decision Tree
BNSP Associate Data Scientist Certification Project

- Processed and prepared a dataset of 766 records through data cleaning, handling invalid values, and preprocessing.
- Developed a Decision Tree classification model using an 80:20 train-test split and class balancing.
- Evaluated model performance on test data, achieving 94.81% accuracy.
- Analyzed feature importance and confusion matrix to assess model performance.

Clothing Classification | Python, Scikit-learn, OpenCV, SVM

- Developed an image classification model to classify clothing into five categories using 3,203 images.
- Extracted image features using Histogram of Oriented Gradients (HOG) and RGB color histograms.
- Applied feature scaling and Support Vector Machine (SVM) with an RBF kernel.
- Achieved 83.46% classification accuracy in the experiment.